Plant Disease Detection System Using Convolutional Neural Networks

Abstract

This project presents a Plant Disease Detection System that uses Convolutional Neural Networks (CNN) to automatically identify plant diseases from leaf images. The system analyzes uploaded leaf photographs and predicts possible diseases with high accuracy. Based on the detected disease, the system provides diagnostic feedback and recommended treatments to farmers and agricultural researchers.

Introduction

Plant diseases significantly affect crop productivity and agricultural sustainability. Traditional disease detection methods rely on manual inspection by experts, which can be time-consuming and inaccessible to many farmers. Computer vision and deep learning technologies enable automated disease detection by analyzing plant images.

Problem Statement

Farmers often struggle to identify plant diseases at an early stage due to lack of expertise or diagnostic tools. Delayed detection can lead to crop damage, reduced yield, and economic losses.

Objectives

The objective of this project is to develop an automated plant disease detection system using CNN models that can analyze leaf images and provide instant diagnostic feedback along with treatment recommendations.

System Architecture

The system includes modules for image input, preprocessing, deep learning model prediction, disease classification, and recommendation generation for treatment.

Dataset

The dataset contains labeled images of healthy and diseased plant leaves. Images are used to train the CNN model to recognize different disease patterns.

Model Implementation

The CNN model is implemented using deep learning frameworks such as TensorFlow or PyTorch. The model processes leaf images through convolutional layers, pooling layers, and fully connected layers to classify plant diseases.

Model Evaluation

The performance of the model is evaluated using metrics such as accuracy, precision, recall, and confusion matrix. The system aims to achieve disease prediction accuracy above 90 percent.

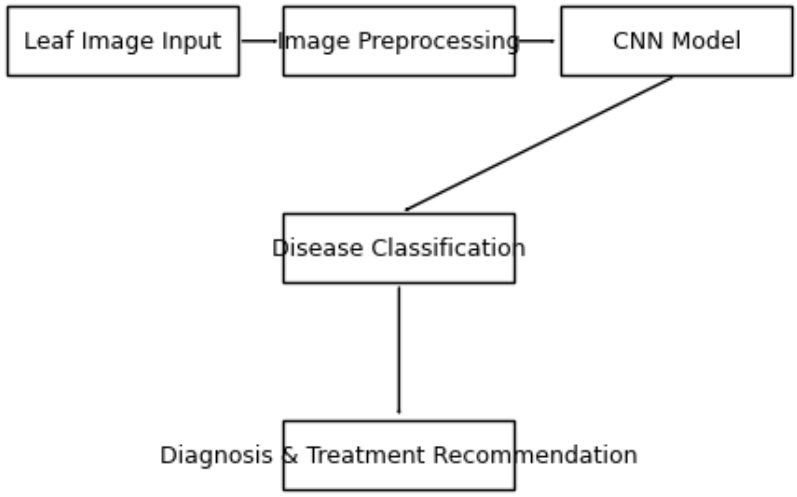
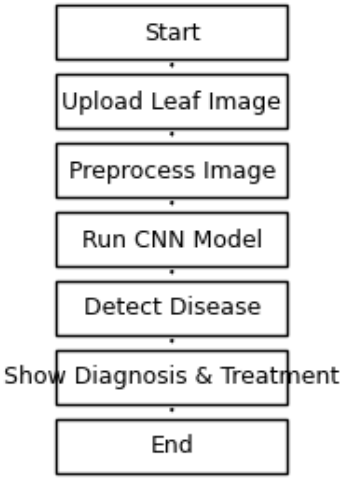
Applications

This system can be used by farmers, agricultural extension workers, and researchers to monitor crop health and take timely action against plant diseases.

Conclusion

The Plant Disease Detection System demonstrates how deep learning can improve agricultural productivity by enabling fast and accurate disease diagnosis. The system supports early detection, reducing crop loss and improving farm management practices.

System Architecture Diagram



Workflow Diagram